

A bounded correction to the covering bound

Disproof of Conjecture 00000008421

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Abstract

For every integer $t \geq 2$, the genuine covering number satisfies $C(t+2, t+1, t) = t+1$. Its ratio to the elementary counting bound is $2(t+1)/(t+2)$, so its relative correction is $t/(t+2) < 1$. This cannot be $\Theta(\log t)$. The counterexamples have strictly interior parameters $t+2 > t+1 > t \geq 2$. The Lean project verifies the covering families, their optimality, the binomial ratio, and the failure of every eventual positive logarithmic lower bound.

The claim and the parameter regime

The source defines $C(n, k, t)$ as the minimum number of k -element blocks needed to cover all t -element subsets of an n -element set. It asserts

$$C(n, k, t) = \frac{\binom{n}{t}}{\binom{k}{t}} (1 + \Theta(\log t)),$$

with the correction stated for $t \geq 2$. A positive $\Theta(\log t)$ correction requires, in particular, a constant $c > 0$ and a threshold beyond which the relative correction is at least $c \log t$. We disprove even that necessary lower bound along the admissible infinite family $n = t+2, k = t+1, t \rightarrow \infty$. The source specifies no additional growth restriction excluding this family. No claim about a different, restricted asymptotic regime is needed.

The exact covering number

Theorem 1. *For every integer $t \geq 0$,*

$$C(t+2, t+1, t) = t+1.$$

Proof. Let V be a set of $t+2$ vertices. Every $(t+1)$ -element block is uniquely $B_v = V \setminus \{v\}$ for some $v \in V$. Thus any family of distinct blocks corresponds bijectively to a set $S \subseteq V$ of chosen omitted vertices; the number of blocks is $|S|$.

Each t -element subset is $V \setminus \{a, b\}$ for two distinct vertices a, b . It is contained in B_v exactly when $v \in \{a, b\}$. Consequently the family covers every t -subset if and only if S meets every pair of distinct vertices. This holds exactly when $|V \setminus S| \leq 1$: two omitted choices would give an uncovered pair, while a set missing at most one vertex meets every pair. It follows that every cover has at least $t+1$ blocks. Choosing all omitted vertices except one gives a cover with exactly $t+1$ blocks. Repeated blocks cannot improve a minimum, so considering distinct blocks loses no generality. $\square$

Failure of the proposed correction

The counting bound for this family is

$$L_t = \frac{\binom{t+2}{t}}{\binom{t+1}{t}} = \frac{(t+2)(t+1)/2}{t+1} = \frac{t+2}{2}.$$

Therefore its relative correction is exactly

$$E_t = \frac{C(t+2, t+1, t)}{L_t} - 1 = \frac{2(t+1)}{t+2} - 1 = \frac{t}{t+2} < 1.$$

For every $c > 0$ and every threshold T , there exists an integer $t \geq \max\{T, 2\}$ such that $c \log t > 1$, because $\log t \rightarrow \infty$. For that t , $E_t < 1 < c \log t$. Hence no eventual lower estimate $E_t \geq c \log t$ with $c > 0$ can hold. In particular E_t is not $\Theta(\log t)$, which disproves the displayed equality in the conjecture. The additional description of that correction cannot restore the failed equality.

Formal correspondence and reproduction

The predicate `Covers` in `lean/Main.lean` uses actual finite families of subsets of `Fin N`: each block has the specified size, and every subset of the target size must be contained in a block. The function `coveringNumber` is the infimum in $\mathbb{N}$ of the cardinalities of those actual covering families. The proof provides a feasible family before identifying this infimum.

The theorem `block_representation` proves the unique omitted-vertex representation of every block. The lower bound `cover_lower_bound` applies to every covering family, not only to the displayed construction. The explicit family `optimalFamily` supplies the matching upper bound. The resulting theorem `coveringNumber_exact` computes the minimum, while `countingBound_exact` proves the actual binomial ratio. The theorems `excess_exact` and `no_eventual_log_lower_bound` establish the relative correction and the quantified contradiction for every $c > 0$ and every threshold. The final theorem is `conjecture8421_false`. Lean’s variable `n` is the paper’s parameter t , so its ambient vertex count is `n+2`.

The project is pinned to Lean 4.19.0 and a fixed Mathlib commit. From `lean/`, run `lake build`, then `lake env lean -DwarningAsError=true Main.lean`. On a new machine, `lake exe cache get` retrieves the official pinned dependency artifacts. The submitted proof itself is verified in a fresh build directory. Printed theorem dependencies use only the standard `propext`, `Classical.choice`, and `Quot.sound`; there are no added axioms, proof placeholders, or numerical oracles. Run `tectonic main.tex` to reproduce the PDF. No auxiliary numerical script is required.

Source

The Last Math Competition, Conjecture 00000008421. The exact bilingual source, its commit, and its SHA-256 are preserved in the submission.